MindBridge Automation Strategy

Project: MindBridge AI Tutor
Sprint: Sprint 1
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Purpose

This document identifies the highest-value test cases for automation in the MindBridge AI Tutor application and recommends at least one test that should remain manual. The recommendations are based on automation best practices, such as stability, repeatability, predictable outcomes, and critical business workflows, as well as the expected value, ROI, and priority of automating the feature versus the effort required.

Section 1—Top 3 Automation Candidates

Automation Candidate 1

- **Test Case ID:TC-01**
- **Test Case Name:** Verify that a student can log in to the MindBridge AI Tutor application using a valid email and password on the QA environment (localhost:8081).
- **Reason for Automation (1–2 sentences):** Since this critical authentication path is repeated in every build, it is highly repeatable and possesses a stable nature.
- **Expected Automation Value / Priority (for example, execution frequency, estimated time saved):**It is executed in every build, saving approximately 2 minutes per test cycle.High
- **Automation Characteristics (check all that apply):**
    - [x] Highly repetitive
    - [x] Stable feature
    - [x] Critical business path
    - [x] Predictable outcome
    - ☐ Data-driven
    - [x] Runs on every build
    - [x] Regression test
    - ☐ Other: _____

Automation Candidate 2

- **Test Case ID:** TC-03
- **Test Case Name:** Verify that the Course Registration checkout flow successfully processes a mock payment and confirms enrollment for a student.
- **Reason for Automation (1–2 sentences):**This is the application's core business process (critical path) and must be reliably verified before every release.

- **Expected Automation Value / Priority (for example, execution frequency, estimated time saved):** As it constitutes core functionality, it is a high priority and prevents manual errors.Critical
- **Automation Characteristics (check all that apply):**
    - ☐ Highly repetitive
    - ☐ Stable feature
    - [x] Critical business path
    - [x] Predictable outcome
    - ☐ Data-driven
    - [x] Runs on every build
    - [x] Regression test
    - ☐ Other: _____

Automation Candidate 3

- **Test Case ID:** TC-07
- **Test Case Name:** Verify that bulk importing 200 new student records from a CSV file into the MindBridge system results in all records being accurately created in the database.
- **Reason for Automation (1–2 sentences):** Manually entering and verifying 200 rows of data is time-consuming and prone to errors, whereas data-driven automation performs this task quickly and flawlessly.
- **Expected Automation Value / Priority (for example, execution frequency, estimated time saved):** It saves 10–15 minutes of manual operation time for each large data upload.Medium
- **Automation Characteristics (check all that apply):**
    - [x] Highly repetitive
    - ☐ Stable feature
    - ☐ Critical business path
    - [x] Predictable outcome
    - [x] Data-driven
    - ☐ Runs on every build
    - [x] Regression test
    - ☐ Other: _____

Section 2—Recommended Manual Test

- **Test Case ID:** TC-05
- **Test Case Name:**Explore the new AI Practice Generator feature for unexpected usability issues, confusing workflows, or edge-case bugs under slow network conditions.
- **Reason this test should remain manual (1–2 sentences):** Exploratory testing and user experience (UX) evaluation are not suitable for automation because they require subjective human judgment.
- **Why Manual Testing is Preferred (check all that apply):**
    - [x] Requires human judgment
    - [x] User experience (UX)
    - [x] Exploratory testing

- ☐ Visual assessment
- ☐ Accessibility evaluation
- ☐ Feature still evolving
- [x] Edge-case investigation
- ☐ Other: _____

Automation Strategy Summary

- Why these tests were selected for automation (including value/priority justification): These tests were selected because they are repetitive, time-consuming, and cover critical business paths. Automation ensures the accurate and rapid execution of tests with every release, eliminating human error and freeing up the testing team to focus on high-value, creative tasks.
- Why the selected manual test should not be automated: Since the selected test primarily examines user experience (UX), visual appearance, and exploratory-type errors—areas that require human judgment and subjective evaluation—it is not suitable for automation via scripts.

Reviewer (Optional)

Reviewed By: Gültəkin Əzizova
Date:2026-10-17

Submission Checklist

Before exporting this document as a PDF, confirm:

- [x] Exactly three automation candidates are identified.
- [x] At least one manual test is identified.
- [x] Each selection includes a clear justification.
- [x] Each automation candidate includes a defined expected value or priority metric (for example, frequency, time saved) to justify the automation effort.
- [x] Automation justifications reference concepts such as stability, repeatability, predictable outcomes, data-driven testing, critical path, or regression testing.
- [x] Manual test justification references concepts such as UX, exploratory testing, accessibility, visual assessment, or human judgment.
- [x] The document has been saved and exported as a PDF before submission.